

H2GUARD · CALCULATION NOTE

H2Guard sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); budget \$265, R13 scope, timed escalation (F7, F8) and head placement basis (E7) added; status table updated
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (sources and inventory, build-up, dilution, plume at the head, response chain, arrestor lag, fan and duct system, power, bump test, alarm level, placement, installation time, cost, trip chain FMEA)

H2Guard sizing calculations

On paper, H2Guard now meets eleven of its fifteen requirements (seven by calculation, four by design), has three at risk and one that cannot be verified at TRL 3; none is missed outright. Amish accepted the review recommendations on 2026-09-25 (HGD-DDR-002). **R14 (cost) is met on paper:** the priced BOM of \$264 sits \$1 under the new \$265 `budget_usd` (it was 47 % over \$180). **R13 (installation) moves from not met to at risk:** with the wall openings now builder's work outside the target, the estimate is 4.1 h against 4 h. R4 (response time) is at risk because the catalytic sensor's t_{90} is still an assumed 30 s, and R7 (ventilation) is at risk because the fan curve is assumed. R1's resolution cannot be checked without sensor data. The calculations changed three parts of the TRL 2 concept: the 300 m³/h axial fan cannot deliver the 300 m³/h boost against the duct and grille losses (237 m³/h), so it is re-specified as a 450 m³/h mixed-flow fan; the 5 mm flame arrestor disc with the sensor 15 mm behind it would add about 21 s of diffusion lag, so the disc is 2 mm thick with the sensor about 2 mm behind it (1.5 s); and a bump test cup, tube and port are added so that R12 can be met. The design leak trips the system at once only if the detector head sits on the plume axis; off the axis the head sees about 15 % LFL and warns. The accepted timed escalation now closes the supply when a warning is held for 5 min, at about 5.6 min for the off-axis design leak with 28 L released (Section F). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [E2], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that a room is safe, and H2Guard is a research and teaching prototype, not a certified gas detection system. Hydrogen is flammable in air from about 4 % to 74 % by volume. See HGD-PRC-001, Safety.

■ Scope and method

The note checks every requirement in HGD-REQ-001 v0.4 against the design in HGD-PRC-001 v0.4 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS` and derived dimensions, so the room, mounting heights, sensor cavity, bump test cup and tube are those in the STEP files and in drawing HGD-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.txt`.

The design case is the 30 m³ reference room (4 x 3 x 2.5 m) with a 5 L/min hydrogen leak from fittings on the bench apparatus, 1,150 mm above the floor, with the detector head directly above it.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Hydrogen	LFL 4.0 % vol; 20 °C, 1 atm; molar volume 24.05 L/mol; density 0.0838 kg/m ³ (air 1.204)	US DOE fact sheet cited in HGD-PRB-001; ideal gas
Leak	5 L/min design leak downstream of a flow restrictor; point source at the top of the apparatus	HGD-REQ-001
Plume	Morton, Taylor and Turner point-source plume, top-hat entrainment coefficient 0.10 (0.08 to 0.12 checked); axis value twice the top-hat mean; leak momentum and non-Boussinesq effects near the source ignored	Textbook plume theory; the factor of two is an assumption
Ventilation	Exhaust at high level, make-up air at low level on the far side; general ventilation of the room taken as zero	HGD-REQ-001
Duct losses	Loss coefficients on the 150 mm duct velocity: inside grille 0.7, backdraft shutter 1.2, weather hood with exit 1.5, sleeve friction factor 0.03; make-up grille K 2.0 on 50 % free area	Handbook ranges; to confirm with the chosen parts
Fan curves	Quadratic, $p = p_0 (1 - (Q/Q_{\max})^2)$. TRL 2 axial fan: $Q_{\max}$ 300 m ³ /h, p_0 100 Pa. TRL 3 mixed-flow fan: $Q_{\max}$ 450 m ³ /h, p_0 200 Pa, 35 W input at full speed; power scales with speed cubed	Typical 150 mm fans; no datasheet chosen yet
Sensor	Catalytic sensor t_{90} 30 s (target, not confirmed); power 0.5 W; MOS sensor 0.28 W	TRL 2 figures; the Figaro page fetched on 2026-09-25 says only "fast response"
Arrestor	Sintered stainless disc, porosity 0.40, tortuosity 3; hydrogen diffusivity in air 0.61 cm ² /s at 0 °C, scaled by $T^{1.75}$	Handbook values
Trip chain	Comparator 0.05 s, relay drop-out 0.02 s, valve closing 0.10 s with a Zener clamp on the coil; watchdog 1.0 s	Typical small parts; to confirm
Other loads	Controller and display 0.5 W; valve coil 8 W; sounder and beacon 3 W	TRL 2 figures
Bump test	Span gas 1 % vol hydrogen at 1.0 L/min; cup flushed by five volumes; 34 L disposable cylinder	Common practice; cup delivery factor not known
Installation	Task times in Section M are estimates for one competent person with hand tools and a core drill	Judgment; to confirm in use

■ A. Set points

- Warning at 10 % LFL is 0.40 % vol; trip at 25 % LFL is 1.00 % vol [A1]. The reference room is 30.0 m³ with a 12.0 m² ceiling [A2].

■ B. Sources and inventory (R8, R9)

- **Electrolyzer.** A 100 W PEM stack at 1.9 V per cell makes 2.73×10^{-4} mol/s, 0.39 L/min. The 5 L/min design leak is 13 times this [B1].
- **Inventory limit.** One percent of the room volume is 300 L at 1 atm, 25.1 g of hydrogen [B2]. H2Bench's 2 L tank at 300 kPa gauge holds 7.9 L, 2.6 % of the limit; a 10 L cylinder at 200 bar holds 1.75 m³, 5.8 times the limit [B3].

- **Flow restrictor.** A choked orifice of 0.13 mm at 10 bar gauge, or 0.21 mm at 3 bar gauge, limits a full-bore failure downstream to the design leak [B4]. This sizes the restrictor in the gas system; it is not H2Guard hardware.

■ C. Build-up without ventilation

Table 2. Time to reach the set points with a 5 L/min leak and no ventilation [C1, C2].

Level	Ceiling layer 0.3 m (3.6 m ³)	Time	Whole room mixed	Time
10 % LFL	14 L	2.9 min	120 L	24 min
25 % LFL	36 L	7.2 min	300 L	60 min
100 % LFL	144 L	28.8 min	1,200 L	240 min

The TRL 2 figures stand.

■ D. Dilution with ventilation (R8)

- **Well mixed.** At 150 m³/h the steady concentration for the design leak is 0.200 % vol (5.0 % LFL); at 300 m³/h it is 0.100 % vol (2.5 % LFL). The electrolyzer case gives 0.0157 % vol [D1]. The room time constant at 5 air changes per hour is 12 min [D2].
- **Stratified.** With exhaust at high level, all the hydrogen leaves through the fan at steady state, so the upper layer settles at the exhaust value, 5.0 % LFL [D3]. The plume would carry the full exhaust flow only 2.28 m above the source, and the ceiling is 1.35 m above it, so no clean interface forms: the room above source height is at about the exhaust value [D4]. The TRL 2 worry that the ceiling layer could be five times richer than the room average does not hold at steady state; it does hold near the plume (Section E) and in the first minutes of a leak.

■ E. Plume at the detector head (R2, R3)

- **Plume.** The design leak has a buoyancy flux of 7.61×10^{-4} m⁴/s³; the sensor ports are 1.175 m above the source [E1]. At the head the plume carries 13.8 L/s of mixture at a mean of 0.60 % vol (15 % LFL), and about 1.21 % vol (30 % LFL) on its axis [E2]. It rises at 0.22 m/s and takes 4.0 s to reach the head [E3].
- **What the head sees.** A head on the plume axis reads above the trip; a head beside the axis reads about the mean and warns but does not trip [E4]. Leaks of 3.8 L/min or more reach 25 % LFL on the axis; leaks of 2.7 L/min or more bring the plume mean to 10 % LFL [E5]. An entrainment coefficient between 0.08 and 0.12 moves the axis value to between 24 and 41 % LFL [E6]. The plume's top-hat radius at the ports is about 141 mm [E7], which is why the head must sit over the leak point.
- **Consequence.** For the design leak with the fan running, the room stays at about 5 % LFL (R8 met), but whether the supply closes depends on the head being on the plume axis. A smaller leak, or a leak from a fitting away from the head, is warned but not stopped by the set points alone. Amish accepted two remedies on 2026-09-25 (HGD-DDR-002): a timed escalation that closes the valve when a warning is held for 5 min (Section F, [F7]), and a placement rule that keeps the head directly above each likely leak point, with a second head where leak points are far apart. The horizontal offset limit is to be set at TRL 4 from a measured plume width; the 141 mm radius above is the paper basis.

■ F. Response chain (R4, R5, R6)

- **Flame arrestor lag.** A 25 x 2 mm sintered disc has an effective diffusivity of 9.2×10^{-6} m²/s and a slab time of 0.43 s. The 1.08 mL cavity between disc and sensor can reaches 90 % in 1.10 s, 1.5 s in all [F1]. The TRL 2 arrangement, a 5 mm disc with the sensor 15 mm behind it, would have added about 21 s [F2], using most of the R4 budget, so the disc is thinned and the sensor moved close to it (`PARAMS["disc"]` , `PARAMS["board_gap"]`).
- **Leak to valve closed.** Rise 4.0 s, arrestor 1.5 s, sensor t90 30 s (assumed), comparator, relay and valve 0.17 s: 35.7 s [F3]. Trip to valve closed is 0.17 s against the 2 s limit, and the alarm follows a step at the head within 32 s against the 60 s limit [F4].
- **Gas released.** About 3.0 L escapes at 5 L/min before the valve closes, plus 0.47 L held in 3 m of 4.3 mm bore line at 10 bar gauge downstream of it [F5]. Both are 1 % of the inventory limit or less.
- **Faults.** A watchdog timeout closes the valve in 1.12 s; a sensor open or short circuit through the comparator in 0.17 s; a stopped fan, once detected after 10 s without tachometer pulses, in 0.12 s [F6]. Loss of 24 V closes the valve as the coil current decays.
- **Timed escalation (HGD-DDR-002).** A warning held for 300 s closes the valve through the microcontroller and latches like a trip. For the off-axis design leak the warning comes at 35.5 s and the valve closes at 336 s (5.6 min), with 28 L released, 9 % of the R9 inventory limit; at the 2.7 L/min warning threshold leak about 15 L is released [F7]. With the fan running the well-mixed room is at about 1.9 % LFL at that moment, against a steady 5.0 % LFL [F8]. The rule is a firmware rule, so it does not share the independence of the comparator trip (R6); leaks below 2.7 L/min at the head give no warning and are not escalated [F8].

■ G. Fan and duct system (R7)

- **Losses.** At 150 m³/h the duct velocity is 2.36 m/s; the exhaust side has a total loss coefficient of 3.44, and the make-up grille has 0.024 m² of free area [G1]. The system needs 15.1 Pa at 150 m³/h and 60.5 Pa at 300 m³/h [G2].
- **TRL 2 fan fails the boost.** A 150 mm axial fan rated 300 m³/h in free air meets the losses at 237 m³/h, 7.9 air changes per hour, 21 % short of the boost target. A fan rated at the target flow in free air can never deliver it through a real duct [G3].
- **Re-specified fan.** A 150 mm mixed-flow EC fan of about 450 m³/h free air and 200 Pa shut-off gives 347 m³/h at full speed, 11.6 air changes per hour, 16 % above the boost target [G3]. The continuous 150 m³/h needs 43 % speed and about 3.8 W; boost takes about 36 W [G4]. At boost the room runs about 19 Pa below ambient across the make-up grille [G5], which a door undercut will relieve in practice. R7 stays at risk until a real fan curve is used.

■ H. Power (R11)

Table 3. Power by state [H1].

State	Power
Normal (valve open, fan continuous)	13.1 W
Warning (valve open, fan boost, intermittent sounder)	48.3 W
Trip (valve closed, fan boost, sounder and beacon)	40.3 W

The peak of 48.3 W leaves a margin of 1.24 on the 60 W supply; the valve coil dissipates 8 W continuously, and the normal state uses 0.31 kWh per day [H2]. The TRL 2 figures (21 W normal, 29 W alarm) are replaced: normal falls because the larger fan runs slower, and the warning state now counts the valve and fan boost together.

■ J. Bump test and log (R12)

- **Bump test.** Span gas at 1.0 L/min passes through 2.69 m of tube (13.2 mL, 0.8 s) into the 114 mL cup under the head, which five volumes flush in 34 s; with the arrestor and the assumed sensor t90 the reading settles after about 67 s, within the 120 s limit [J1]. Each test uses 1.36 L of span gas, so a 34 L disposable cylinder gives about 25 tests [J2]. The cup is open at the bottom, so the concentration the sensor sees is lower than the span gas by an unknown delivery factor; the pass level must be set when that factor is measured.
- **Trip during the test.** Span gas of 1 % vol equals the trip level, so a bump test that reaches full span trips the system. This proves the whole chain, but the valve then needs a key reset; whether to keep this or use a lower span gas remains open for Amish.
- **Log.** One 16 B record per minute for 90 days is 2.07 MB plus events, so the board needs 4 MB of flash or more [J3]. The BOM now says so.

■ K. Alarm level (R10)

- A sounder of 90 dB at 1 m gives 80 dB at 3 m and 76 dB at 5 m in free field [K1]. The farthest point of the room is 5.0 m away, where the sounder is 16 dB above a 60 dB lab background [K2].

■ L. Placement and ignition sources (R1, R15)

- The sensor ports are 175 mm below the ceiling (R1: 300 mm or less); the fan grille's top is 80 mm below it [L1].
- The controller's top is 1,025 mm below the ceiling (R15: 1,000 mm or more), the beacon's top 450 mm, and the power supply is on the floor [L2].
- The arrestor pore size of 50 µm or less is well below the maximum experimental safe gap for hydrogen, about 0.3 mm (IEC group IIC; value to be checked against IEC 60079-20-1). The catalytic element runs hot, and no part is rated for hazardous areas.

■ M. Installation time (R13)

The estimate is 360 min (6.0 h) with both wall openings made on the day, and 245 min (4.1 h) if a builder makes them beforehand [M1]. The 160 mm core through the wall (90 min) and the make-up air opening (60 min) dominate. Under HGD-DDR-002 the fan and grille wall openings are builder's work outside the R13 target, so the 4.1 h figure applies: 5 min over the 4 h target on a judgment estimate, so R13 is at risk rather than met.

■ N. Cost (R14)

The BOM has 15 lines totaling \$264.00, \$1.00 under the \$265 `budget_usd` set by Amish's acceptance of the recommendation (HGD-DDR-002; it was \$180, 47 % short) [N1]. The margin is small, and the prices are indicative. Without the fan and make-up grille (\$58) the kit is \$206 [N2]. The TRL 3 changes added \$15 (larger fan), \$9 (bump test port and tube) and \$1 (arrestor and cup), against the TRL 2 total of \$239.

■ P. Failure modes of the trip chain (R5, R6)

Table 4. Failure modes and effects of the trip chain (first pass).

Failure	Effect	Detected by	Result
Loss of 24 V	Valve coil de-energized	Inherent	Safe: valve closes
Microcontroller stops	Valve MOSFET off or watchdog reset	1 s watchdog	Safe: valve closes [F6]
Firmware bug holds valve MOSFET on	Comparator relay in series still opens at trip	Comparator	Safe for gas above the trip
Comparator relay contact welds	MOSFET in series still opens on firmware trip	Test button (proof test)	Safe while firmware runs; latent single fault
Valve MOSFET fails short	Comparator relay in series still opens	Test button	Safe; latent single fault
Sensor open or short, heater failure	Bridge outside the comparator window	Comparator	Safe: valve closes [F6]
Fan stops	No tachometer pulses	Controller, 10 s	Safe: valve closes [F6]
Duct blocked, shutter stuck, fan running	Tachometer still pulses	Not detected	Dangerous, undetected: no airflow proof
Sensor poisoned (silicones, sulfur)	Reads low	Bump test only	Dangerous, undetected between tests
Port blocked (dust, paint, cap left on)	Reads low	Bump test only	Dangerous, undetected between tests
Valve seat leaks or sticks open	Supply not stopped	Not detected	Dangerous, undetected: no position or leak check

R5's listed faults all close the valve within 2 s, and R6 holds because the comparator's relay sits in series with the microcontroller's valve switch. Four dangerous faults remain undetected in operation. Two are covered by the bump test before each session. A blocked duct needs airflow proving, which is an open question for Amish. A leaking valve seat needs a proof test that is TRL 4 work.

■ Results against every requirement

Table 5. Requirement status from this note.

ID	Requirement	Value	Target	Status
R13	Install simply	4.1 h with the wall openings as builder's work; 6.0 h with them [M1]	4 h, wall openings excluded (HGD-DDR-002)	At risk (5 min over on estimate)
R4	Respond fast enough	35.7 s leak to valve closed with t90 assumed 30 s; 0.17 s trip to closed [F3, F4]	t90 30 s or less; alarm 60 s; valve 2 s	At risk (t90 unconfirmed)
R7	Ventilate the room	347 m3/h boost, 150 m3/h continuous at 43 % speed; TRL 2 fan 237 m3/h [G3, G4]	150 m3/h; boost 300 m3/h	At risk (fan curve assumed)
R1	Measure at the high point	Ports 175 mm below the ceiling [L1]; plume radius 141 mm at the ports [E7]; 0 to 100 % LFL by selection	0 to 100 % LFL, 1 % LFL resolution, within 0.3 m of the ceiling, directly above each likely leak point	Not verifiable at TRL 3 (resolution needs sensor data); placement met
R2	Warn early	Design leak plume at the head 15 % LFL mean [E2]	Warn at 10 % LFL	Met on paper
R5	Fail safe	1.12 s worst listed fault to closed [F6]	2 s	Met on paper; four dangerous undetected faults (Table 4)
R8	Keep the design leak below the trip	5.0 % LFL well mixed and in the upper layer [D1, D3]	Below 10 % LFL	Met on paper
R10	Alert people	76 dB at 5 m, 16 dB over background [K2]	85 dB(A) at 1 m; beacon visible	Met on paper
R11	Keep mains out; 60 W or less	48.3 W peak [H2]	60 W	Met on paper
R12	Be checkable in use	Bump test 67 s [J1]; log 2.07 MB in 4 MB [J3]	2 min; 90 days	Met on paper (cup delivery factor unknown)
R14	Stay within the concept budget	\$264.00 [N1]	\$265 (budget_usd , HGD-DDR-002)	Met on paper (\$1 margin, indicative prices)
R3	Trip and shut off	Logic and latch; design leak 30 % LFL on the plume axis [E2]; off the axis the 5 min escalation closes the valve at 5.6 min [F7]	Trip at 25 % LFL or a warning held 5 min, key reset	Met by design (escalation is a firmware rule)
R6	Trip without firmware	Comparator relay in series with the valve MOSFET, hardware latch	Independent of the microcontroller	Met by design
R9	Limit what can leak	300 L limit; H2Bench tank 2.6 % of it; 0.13 mm restrictor [B2 to B4]	1 % of room volume or restrictor	Met by design (installation rule)
R15	Avoid ignition sources in the ceiling layer	Controller 1,025 mm below the ceiling; continuous fan; 50 µm arrestors [L2]	1 m; arrestors; continuous fan	Met by design; not certified

Counts: 0 not met, 3 at risk, 1 not verifiable at TRL 3, 7 met on paper, 4 met by design. At v0.1 they were 2 not met (R13, R14), 2 at risk, 1 not verifiable, 6 met on paper and 4 met by design.

■ Checks against the TRL 2 figures

TRL 2 claim (HGD-PRC-001 v0.2)	This note	Action
Electrolyzer about 0.4 L/min	0.39 L/min	Stands
Layer to 25 % LFL about 7 min; room about 60 min	7.2 min; 60 min	Stands
About 5 % LFL well mixed at 150 m3/h	5.0 % LFL	Stands
Ceiling layer could be five times the room average and trip	At steady state the layer is at the exhaust value; the plume at the head is 15 to 30 % LFL	Precis updated
Transport to the head up to 5 s	4.0 s	Stands
Leak to valve closed about 35 s	35.7 s, but only with the arrestor changed; 55 s with the TRL 2 arrestor	Arrestor changed, precis updated
Fan about 300 m3/h free air meets the 300 m3/h boost	237 m3/h	Fan re-specified, precis and BOM updated
About 3 L released; under 1 L in the line	3.0 L; 0.47 L	Stands
Inventory limit 300 L, about 25 g; cylinder more than five times	300 L, 25.1 g; 5.8 times	Stands
Power about 21 W normal, 29 W alarm	13.1 W normal, 48.3 W warning, 40.3 W trip	Precis updated
Parts about \$239	\$264.00	Precis, README and BOM notes updated